

# SEONGTAE BANG

GPU-Efficient Heterogeneous Systems · AI Training Systems · GPU Memory Systems

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## RESEARCH PROFILE

Ph.D. researcher in computer architecture and systems focused on **GPU-efficient heterogeneous systems for memory-intensive AI workloads**.

I identify computation and state that do not need to occupy GPU resources, then restructure execution so that they progress concurrently on CPUs or other system resources. This frees HBM capacity and GPU cycles for highly parallel, accelerator-critical work, improving memory efficiency, GPU utilization, and end-to-end throughput.

A recurring theme is finding what existing systems overlook and turning it into performance: host-memory traffic, not CPU arithmetic, behind slow offloaded optimizers, and GPU cores stalled by page faults that can reclaim memory themselves. My work spans multi-GPU training frameworks, NVIDIA GPU system software, and GPU architecture research. I have modified DeepSpeed and Megatron-LM, redesigned NVIDIA UVM mechanisms at the driver level, and proposed new GPU memory-management architectures through simulation, with first-author papers at **ASPLOS 2027** and **MICRO 2026**.

## SELECTED RESEARCH & ENGINEERING

### ReFlow — GPU-Efficient LLM Training through CPU-GPU Parallelism

First author · ASPLOS 2027 (to appear)

- Identified the exposed host update in CPU-offloaded training as a host-memory-traffic problem rather than a CPU-arithmetic one: non-arithmetic overhead is about **5×** the Adam compute time, and the GPU idles for up to **47.9%** of each iteration on a B200 system.
- Unlike prior systems that hide this delay by moving work back to the GPU or speculating, removed it at the source: register-resident SIMD fusion, BF16 gradient transfer with CPU-side accumulation, and deadline-aware scheduling, preserving exact training semantics (bit-identical losses).
- Reduces GPU memory consumption while achieving throughput comparable to or higher than GPU-resident ZeRO-3 across evaluated configurations (**98%** at 50B on 8 B200 GPUs; **1.2×** higher at 80B). Achieves up to **4×** the throughput of ZeRO-Infinity and **3×** that of SuperOffload, and scales to **95B** parameters on 8 B200 GPUs (ZeRO-Infinity: 80B) with about 20% lower host-memory footprint.
- Builds on ReplayOpt (IEEE CAL 2026, first author), which first separated the parameters the GPU needs now from the FP32 state that can wait, cutting CPU optimizer time by up to **55.2%** and step time by up to **21.7%**.
- Implemented end-to-end in a DeepSpeed fork (framework runtime, gradient transfer, CPU optimizer kernels, multi-GPU execution), released as a [public artifact](#) (Apache-2.0); an upstream pull request to DeepSpeed is in preparation.

### ReclaimX — GPU Architecture Redesign for UVM Oversubscription

First author · MICRO 2026 (to appear)

- Observed that a far-page fault both signals that HBM must be freed and stalls a TPC that can no longer advance foreground work, so the fault itself creates the idle compute needed to relieve memory pressure.
- Unlike prior work that frees memory from the host at coarse 2 MB granularity or throttles and reserves cores, designed reclamation that runs on fault-stalled TPCs: 64 KB victim ordering, selective compressed retention in HBM, and local restore on the GPU.
- Validated with GPU simulation and UVM modeling: **2.33×** geometric-mean speedup over baseline UVM across 12 oversubscribed workloads; with prefetching, **90.6%** fewer far-page faults and up to **3.61×**.

### SAFE & ARIADNE — NVIDIA UVM Driver Redesign

Co-first author · CAL 2025 / Second author · HPCA 2026

- Performed code-level analysis of NVIDIA's open-source UVM driver (page faults, migration, prefetching, sharing, oversubscription policies) and modified it directly rather than treating it as a black box.
- SAFE: showed that how widely a block is shared across SMs predicts whether prefetching pays off, and chose prefetching per block from uTLB-derived sharing status with no hardware changes; up to **6.5×** (**3.6×** average) over the default UVM prefetcher.
- ARIADNE: made UVM robust under heavy oversubscription from the driver alone (pipelined fault handling, runtime Sharing Degree, adaptive GPU-memory vs. zero-copy placement); **1.9×**, **5.0×**, and **4.8×** average speedups over a state-of-the-art method at 130/175/300% oversubscription.

### Memory-Scalable MoE Training

Ongoing

- Extending Megatron-LM with heterogeneous CPU-GPU execution that offloads optimizer and state-management work not requiring GPU resources, while keeping grouped expert GEMMs on GPUs.
- Aims to free HBM so that TP/PP/DP configurations are chosen for throughput rather than memory feasibility, increasing trainable model capacity.

### Memory-Scalable 3D Gaussian Splatting Training

Ongoing

- Pipelining CPU-side culling for upcoming views with current-view GPU rasterization and backward propagation.
- Keeps only the active Gaussian working set in HBM and persistent training state in host memory, aiming to train beyond practical HBM capacity without changing the training objective.

SYSTEMS DEVELOPMENT

**DeepSpeed** — Modified LLM training and optimizer-offload paths (optimizer execution, CPU–GPU synchronization, gradient movement, mixed precision) on multi-GPU systems.

**Megatron-LM** — Modified MoE training paths (expert execution, optimizer/state placement, CPU offloading); analyzed how TP/PP/DP/EP choices interact with HBM capacity and throughput.

**NVIDIA UVM driver** — Source-level analysis and modification of page-fault handling, migration, prefetching, and oversubscription.

TECHNICAL EXPERTISE

|                       |                                                                                                                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages             | C · C++ · Python · CUDA · Bash                                                                                                                                                                            |
| AI Training Systems   | DeepSpeed · Megatron-LM · PyTorch · LLM Training · Mixture-of-Experts · TP/PP/DP/EP · ZeRO · Optimizer Offload · Mixed Precision · Grouped GEMM                                                           |
| GPU Systems & Drivers | NVIDIA UVM · GPU Driver Internals · Linux Kernel Drivers · Page-Fault Handling · Page Migration · Prefetching · GPU Memory Oversubscription · HBM Working-Set Management · Memory Tiering · NVIDIA nvCOMP |
| Heterogeneous & CPU   | CPU–GPU Co-execution · Workload Partitioning · Overlap and Scheduling · AVX-512 SIMD · OpenMP · NUMA · Multithreaded Optimization · Memory-Layout Optimization                                            |
| Performance Analysis  | Nsight Systems · Nsight Compute · perf · Workload Characterization · Memory-Traffic Analysis                                                                                                              |
| Simulation            | Accel-Sim · GPGPU-Sim · UVMsmart · gem5 · pNet-gem5 · GPU Memory-System Simulation                                                                                                                        |
| Platforms             | NVIDIA B200 · A100 · RTX PRO / RTX A-series · AMD EPYC Multi-Socket Systems · Intel IAA                                                                                                                   |

PUBLICATIONS

**ReFlow: Exposing Parallelism in CPU-Offloaded LLM Training via Register-Resident SIMD Fusion and Decoupled Update Scheduling.** Seongtae Bang, Gyeongseo Park, Ki-Dong Kang, Hyunkyun Shin, Sungju Kim, and Daehoon Kim. *Proceedings of the 32nd ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '27)*, Heraklion, Greece, April 2027 [To appear](#) [\[Artifact\]](#)

**ReclaimX: Device-Side Memory Reclamation via Stalled GPU Execution for UVM Oversubscription.** Seongtae Bang, Hyunkyun Shin, Hyungwon Park, Minh Kim, and Daehoon Kim. *IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2026 [To appear](#)

**ARIADNE: Adaptive UVM Management for Efficient GPU Memory Oversubscription.** Hyunkyun Shin, Seongtae Bang, Hyungwon Park, and Daehoon Kim. *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, Sydney, Australia, pp. 1–15, 2026 [\[Artifact\]](#)

**ReplayOpt: Optimizer-State Replay to Resolve Critical-Path Bottlenecks in Offloaded Training.** Seongtae Bang, Gyeongseo Park, Kyeonghyeon Ryu, and Daehoon Kim. *IEEE Computer Architecture Letters*, vol. 25, no. 1, pp. 142–145, 2026

**SAFE: Sharing-Aware Prefetching for Efficient GPU Memory Management with Unified Virtual Memory.** Hyunkyun Shin\*, Seongtae Bang\*, Hyungwon Park, and Daehoon Kim. *IEEE Computer Architecture Letters*, vol. 24, no. 1, pp. 117–120, 2025

**pNet-gem5: Full-System Simulation with High-Performance Networking Enabled by Parallel Network Packet Processing.** Jongmin Shin, Seongtae Bang, Gyeongseo Park, and Daehoon Kim. *IEEE Computer Architecture Letters*, vol. 24, no. 2, pp. 193–196, 2025 [\[Code\]](#)

\* Equal contribution.

RESEARCH EXPERIENCE

**Graduate Researcher** — CASLAB 2020–Present

- Develop and modify established frameworks (DeepSpeed, Megatron-LM) and GPU system software rather than standalone prototypes, moving to architecture simulation when an idea needs hardware changes.
- Build optimized CPU paths (AVX-512 SIMD, OpenMP, NUMA-aware parallelism, memory-layout optimization), evaluate Intel IAA and NVIDIA nvCOMP compression trade-offs, and characterize systems end-to-end across GPU compute, CPU execution, HBM capacity, host-memory bandwidth, and data movement.

EDUCATION

**DGIST** — Ph.D. Student (Integrated M.S.–Ph.D.), Electrical Engineering and Computer Science 2020–Present  
Advisor: Prof. Daehoon Kim · Expected Graduation: February 2027

**DGIST** — Bachelor of Convergence Engineering 2016–2020